NEET-SS / FET / INI-SS HIGH-YIELD NOTES

Foot & Ankle

1. flatfoot and cavus foot
2. hallux valgus and forefoot
3. Achilles and tendons
4. ankle arthritis and instability
5. diabetic foot and Charcot
6. talus and calcaneus

Flatfoot and cavus foot

Part A: Adult acquired flatfoot (progressive collapsing foot deformity)

1. Terminology and anatomy

- **Adult acquired flatfoot deformity (AAFD)**, renamed **progressive collapsing foot deformity (PCFD)** (2020 consensus): **complex 3D deformity** with **hindfoot valgus, forefoot abduction (talonavicular uncoverage), forefoot varus (supination), medial column instability (collapse of the arch)**, and later **ankle valgus**.
- **Tibialis posterior (TP): strongest invertor; primary dynamic stabiliser of the medial longitudinal arch**; inserts mainly on the **navicular tuberosity** with slips to the cuneiforms, cuboid and metatarsal bases 2-4; **locks the transverse tarsal (Chopart) joint** (TN and CC axes become non-parallel: rigid lever for push-off); **antagonist: peroneus brevis**.
- **Hypovascular zone** of TP tendon: **behind and just distal to the medial malleolus** (degeneration site).
- **Static stabilisers: spring ligament (calcaneonavicular: superomedial part most important)**, **deltoid ligament (deep)**, **plantar fascia (windlass mechanism)**, **talocalcaneal interosseous ligament**.
- **Posterior tibial tendon dysfunction (PTTD)** was the classic cause, but **spring ligament failure** is now recognised as central; TP failure is often secondary.

2. Causes

- **TP tendon degeneration (commonest in middle-aged obese women; hypertension, diabetes, steroid injections, seronegative arthropathy)**.
- **Others: inflammatory arthritis (RA)**, **trauma (TP laceration/rupture, Lisfranc injury)**, **neuropathic (Charcot)**, **tarsal coalition (adult**

presentation), **accessory navicular**, **degenerative TMT/naviculocuneiform arthritis**, **iatrogenic**.

3. Clinical evaluation

- **Medial ankle pain along the TP tendon** (early), **lateral sinus tarsi/subfibular impingement pain** (late: calcaneofibular impingement).
- **"Too many toes" sign** (**forefoot abduction seen from behind**).
- **Single-limb heel rise test**: **inability to perform / no heel inversion on heel rise** (**most sensitive test for TP dysfunction**); **double heel rise** to check flexibility (hindfoot inverts if flexible).
- **Flexibility of the hindfoot (subtalar)** and **forefoot (forefoot varus/supination: compensatory fixed or flexible)**: **assess forefoot position with the hindfoot held in neutral**.
- **Gastrocnemius/Achilles contracture** (**Silfverskiöld test**: almost universal).
- **Medial column instability** (first TMT/naviculocuneiform plantar sag), **deltoid laxity (ankle valgus)**.

4. Imaging

- **Weight-bearing radiographs (AP and lateral foot, AP ankle, hindfoot alignment view (Saltzman))**:
  - **Lateral talo-first metatarsal angle (Meary angle): normal 0 degrees (collinear)**; **over 4 degrees convex plantar = flatfoot** (**mild 4-15, moderate 15-30, severe over 30**),
  - **Calcaneal pitch angle (normal about 20-30 degrees; reduced in flatfoot)**,
  - **AP talonavicular coverage angle (over 7 degrees) / talonavicular uncoverage percentage (over 40%)**: forefoot abduction,
  - **AP talo-first metatarsal angle** (**abduction**),
  - **Medial cuneiform-floor height** (reduced), **sinus tarsi impingement**, **subfibular impingement**.
  - **Ankle: talar tilt (valgus) in the mortise** (deltoid failure).

- **Weight-bearing CT** (**3D, subfibular impingement, middle facet subluxation**: **peritalar subluxation**).
- **MRI** (TP tendon tears/tendinosis, spring ligament), **ultrasound**.

5. Classification

Johnson and Strom (modified by Myerson/Bluman)

Stage	TP tendon	Hindfoot	Treatment
I	**Tenosynovitis/tendinosis; normal length**	**Normal alignment; can single heel rise** (painful)	**Non-operative (immobilisation, NSAIDs, orthoses)**; **tenosynovectomy**
II (A/B/C)	**Elongated/dysfunctional**	**Flexible flatfoot** (**IIA mild abduction; IIB abduction over 30-40% TN uncoverage; IIC fixed forefoot varus**)	**Joint-sparing reconstruction**: **FDL transfer to the navicular + medializing calcaneal osteotomy (MDCO) ± lateral column lengthening (Evans) ± Cotton osteotomy ± gastrocnemius recession ± spring ligament repair**
III	**Ruptured/degenerated**	**Rigid flatfoot (fixed hindfoot valgus)**	**Triple arthrodesis** (or **double/modified: TN + subtalar**)
IV (A/B)		**+ ankle valgus (deltoid failure)**: **IVA flexible (correctable), IVB rigid (tibiotalar arthritis)**	**IVA: deltoid reconstruction + hindfoot correction**; **IVB: tibiotalocalcaneal (TTC) or pantalar arthrodesis, or TAR + hindfoot correction**

PCFD consensus classification (2020)

- **Stage 1 (flexible) or 2 (rigid)**, with **classes A (hindfoot valgus), B (midfoot/forefoot abduction), C (forefoot varus/medial column instability), D (peritalar subluxation/dislocation), E (ankle instability: valgus)**; each scored as flexible or rigid.

6. Treatment details

- **Non-operative (all stages first, mainstay in elderly/low demand): CAM boot or short leg cast (acute tenosynovitis, 4-8 weeks)**, **orthoses (medial arch support + medial heel wedge; UCBL)**, **ankle-foot orthosis (Arizona brace)** for later stages, **physiotherapy (eccentric TP strengthening, Achilles stretching)**, weight loss. **Avoid steroid injection into the TP tendon sheath (rupture)**.
- **Flexible deformity (stage II) procedures:**
  - **FDL transfer to the navicular** (**FDL is in phase with and adjacent to the TP; weaker (about 30%) but adequate when combined with bony correction**; **FHL** alternative; **interconnection (Henry knot) allows the FHL to flex lesser toes**),
  - **Medializing displacement calcaneal osteotomy (MDCO)** (**about 10 mm medial shift**: **converts the Achilles tendon pull into an invertor (medial to the subtalar axis)**; **protects the tendon transfer**; **sural nerve and lateral calcaneal branch at risk; the tibial nerve/tarsal tunnel at risk with large shifts**),
  - **Lateral column lengthening (Evans osteotomy: anterior calcaneus between the anterior and middle facets, about 10-15 mm from the CC joint; or CC distraction arthrodesis)** for **forefoot abduction (TN uncoverage over about 40%)**: **lateral column overload (lateral foot pain: commonest complication)**, **CC arthritis**, **sural nerve injury**, non-union,
  - **Cotton osteotomy (plantarflexion opening-wedge osteotomy of the medial cuneiform)** or **first TMT arthrodesis (Lapidus)** for **residual forefoot varus/medial column instability**,
  - **Spring ligament repair/reconstruction**,
  - **Gastrocnemius recession (Strayer) / Achilles lengthening (Hoke)** for equinus,

    - **Subtalar arthroereisis** (sinus tarsi implant: limited role in adults; mainly paediatric).
- **Rigid deformity (stage III): triple arthrodesis (subtalar, TN, CC)** through **medial and lateral incisions (or single medial incision: protects lateral skin)**; **TN joint is the key joint (fusion of TN alone eliminates almost all hindfoot motion)**; **position: hindfoot about 5 degrees valgus, neutral forefoot rotation (plantigrade), neutral abduction**; **complications: non-union (TN commonest site), malunion, adjacent joint arthritis (ankle, naviculocuneiform)**.
- **Stage IV: deltoid ligament reconstruction** (flexible) or **TTC/pantalar arthrodesis** (rigid with arthritis); **total ankle replacement with hindfoot realignment**.

Part B: Pes cavus

7. Definition and causes

- **Elevated medial longitudinal arch (high arch), often with hindfoot varus**; **forefoot-driven (plantarflexed first ray) or hindfoot-driven**.
- **About two-thirds have an underlying neurological cause**: **Charcot-Marie-Tooth disease (HMSN type I: commonest; autosomal dominant; PMP22 duplication on chromosome 17p)**, **spinal cord pathology (tethered cord, diastematomyelia, syringomyelia, tumour, spinal dysraphism)**, **cerebral palsy**, **poliomyelitis (India: residual polio)**, **Friedreich ataxia**, **stroke**, **muscular dystrophy**. **Also: idiopathic/familial (subtle cavovarus)**, **residual clubfoot**, **trauma (compartment syndrome: Volkmann's ischaemic contracture: claw toes; malunited talar neck fracture)**.
- **Unilateral cavus = spinal cord lesion until proved otherwise: MRI whole spine. Bilateral cavus: think CMT (nerve conduction studies, genetic testing)**.

8. CMT pathomechanics

- **Muscle imbalance (pairs of agonist/antagonist; weakest muscles first: tibialis anterior and peroneus brevis; intrinsics):**
  - **Weak tibialis anterior vs strong peroneus longus**: **plantarflexed first ray (forefoot pronation/valgus)**,
  - **Weak peroneus brevis vs strong tibialis posterior**: **hindfoot varus, forefoot adduction**,
  - **Intrinsic weakness**: **claw toes** (**EDL recruited to dorsiflex the ankle: hyperextends MTPJs**),
  - **Plantar fascia contracture** (windlass: elevates the arch).
- **Tripod effect**: **plantarflexed first ray forces the hindfoot into varus on weight bearing (forefoot-driven hindfoot varus)**.
- **Lateral ankle instability, peroneal tendon tears, fifth metatarsal stress fractures, lateral foot callosities, stress fractures, ankle arthritis (varus)**.

9. Evaluation

- **Peek-a-boo heel sign** (**medial heel pad visible from the front with the feet parallel: hindfoot varus**).
- **Coleman block test** (**block under the lateral foot (heel and lateral forefoot), allowing the first ray to drop off the edge**): **hindfoot varus corrects = flexible hindfoot, deformity is forefoot-driven** (treat the forefoot: **first metatarsal dorsiflexion osteotomy + plantar fascia release**); **does not correct = fixed hindfoot varus (needs calcaneal osteotomy or arthrodesis)**.
- **Muscle power testing** (MRC grading of each muscle: guides tendon transfers), **neurological examination** (reflexes, sensation, spine), **calluses (under first and fifth metatarsal heads)**.
- **Radiographs: Meary angle (talo-first metatarsal) convex dorsally (negative: over -4 degrees = cavus)**, **calcaneal pitch angle increased (over 30 degrees: calcaneocavus)**, **Hibbs angle** (calcaneus axis vs first metatarsal axis: **under 150 degrees** = cavus), **"double talar dome"** (ankle AP/lateral: hindfoot varus/external rotation), **fibula posterior**.

10. Treatment

- **Non-operative: orthoses (lateral forefoot post + recess under the first metatarsal head)**, **lateral heel wedge** (for flexible hindfoot), **AFO** (foot drop), **physiotherapy**, **footwear**.
- **Operative (individualised; "flexible deformities need soft-tissue/tendon procedures, fixed deformities need bony procedures, arthrodesis for severe rigid deformity or arthritis"):**
  - **Soft tissue: plantar fascia release (Steindler)**, **gastrocnemius recession/Achilles lengthening** (**in calcaneocavus (polio/spina bifida) the Achilles is weak: do not lengthen**).
  - **Tendon transfers (balance forces): peroneus longus to peroneus brevis transfer** (**removes plantarflexion of the first ray; augments eversion**), **tibialis posterior transfer to the dorsum (through the interosseous membrane to the lateral cuneiform)** for **foot drop**, **Jones transfer (EHL to the first metatarsal neck + hallux IPJ fusion)** for **clawed hallux**, **Hibbs transfer (EDL to the lateral cuneiform/cuboid)** for **claw lesser toes**, **flexor-to-extensor transfer (Girdlestone-Taylor)** for flexible claw toes.
  - **Bony (joint-sparing): dorsiflexion closing-wedge osteotomy of the first metatarsal base** (**forefoot-driven cavus**), **lateral closing-wedge calcaneal osteotomy (Dwyer)** or **lateralising calcaneal (Z) osteotomy** (**fixed hindfoot varus**), **midfoot osteotomy (Cole: dorsal closing wedge through the navicular-cuneiform/cuboid; Japas V-osteotomy; Akron dome)** for **midfoot cavus**, **anterior tibial osteotomy** rarely.
  - **Arthrodesis: triple arthrodesis** (**rigid deformity, arthritis, salvage**; **worse long-term results in CMT (ankle/midfoot arthritis)**: **avoid when possible in young patients**).
  - **Lateral ankle ligament reconstruction** (Broström) **must be combined with correction of the varus** (otherwise fails).

11. Paediatric flatfoot (brief link)

- **Flexible flatfoot: physiological; no treatment unless painful + tight heel cord (calcaneal lengthening + gastrocnemius recession)**.
- **Rigid flatfoot: tarsal coalition, CVT, accessory navicular, inflammatory, neuromuscular**: see paediatric foot notes.

12. Other conditions

- **Tibialis anterior tendon rupture: elderly men (sixth-seventh decade)**, **degenerative**, **foot slap/steppage gait**, **palpable gap/mass at the anterior ankle**, **EHL/EDL substitution (toe clawing on dorsiflexion)**; **repair/reconstruction (EHL transfer, graft) in active patients**; **AFO in low-demand**.
- **Accessory navicular (os naviculare)** in adults: **Kidner procedure (excision + TP advancement)**.
- **Tarsal tunnel syndrome: tibial nerve compression beneath the flexor retinaculum (laciniate ligament)**; **space-occupying lesions (ganglion, varicosities, lipoma, accessory muscles: flexor digitorum accessorius longus)**, **hindfoot valgus (traction)**, trauma; **burning plantar paraesthesia (sparing the heel (calcaneal branch arises proximal) in some cases), Tinel sign, dorsiflexion-eversion test**; **NCS**; **MRI**; **decompression (release of the flexor retinaculum + abductor hallucis fascia (medial and lateral plantar nerve tunnels))**: **results less predictable than CTS (about 50-75%)**, best with a **space-occupying lesion**.
- **Baxter nerve (first branch of the lateral plantar nerve / inferior calcaneal nerve) entrapment**: **chronic heel pain (medial)** mimicking plantar fasciitis; **abductor digiti minimi atrophy on MRI**; decompression with plantar fascia release.
- **Plantar fasciitis (heel pain): degenerative fasciosis at the medial calcaneal tubercle origin**; **first-step morning pain (post-static dyskinesia)**; **calcaneal spur (in FDB origin) is incidental**; **non-operative (about 90%: plantar fascia-specific stretching (best), Achilles stretching, heel cups, night splints, NSAIDs)**; **steroid**

injection (risk of fat pad atrophy and plantar fascia rupture), **ESWT**; **surgery (partial plantar fascia release (medial third to half: avoid complete release: lateral column overload)** ± **Baxter nerve release, gastrocnemius recession)** after **6-12 months**.

Exam pearl: **PCFD/AAFD: TP + spring ligament failure; too many toes; can't single-heel-rise (most sensitive). Johnson-Strom: I tendinosis (conservative), II flexible (FDL transfer + MDCO ± Evans ± Cotton ± gastroc recession), III rigid (triple arthrodesis), IV ankle valgus (deltoid recon or TTC/TAR). MDCO converts Achilles to an invertor; Evans = lateral column overload.**

Exam pearl: **Meary angle: plantar convex = flatfoot, dorsal convex = cavus. Calcaneal pitch high = calcaneocavus. TN joint key to triple arthrodesis (TN non-union commonest); position 5 degrees valgus.**

Exam pearl: **Cavus: unilateral = spinal MRI; bilateral = CMT (PMP22). CMT: weak TA vs strong PL (plantarflexed first ray), weak PB vs strong TP (varus). Coleman block: corrects = forefoot-driven (first MT dorsiflexion osteotomy + plantar fascia release + PL to PB); does not correct = add Dwyer/lateralising calcaneal osteotomy. Jones transfer for clawed hallux. Broström fails without varus correction.**

Exam pearl: **Tarsal tunnel: flexor retinaculum, space-occupying lesion gives best release results. Plantar fasciitis: stretching best; partial release only. Baxter nerve: ADM atrophy.**

Hallux valgus and forefoot

Part A: Hallux valgus

1. Definition and aetiology

- **Lateral deviation of the hallux at the first MTPJ with medial deviation of the first metatarsal (metatarsus primus varus)**, with a medial prominence ("bunion").
- **Women (about 10:1 in surgical series)**, **familial (about 60-90% of juvenile cases; maternal transmission)**, **constricting footwear (narrow toe box, high heels)**, **ligamentous laxity**, **pes planus**, **first ray hypermobility (debated)**, **long first metatarsal**, **rounded metatarsal head**, **increased DMAA**, **inflammatory arthritis (RA)**, **neuromuscular (cerebral palsy, stroke)**, **amputation of the second toe**.

2. Pathoanatomy (sequence)

1. **Medial capsule and medial sesamoid ligament attenuation**; **the hallux drifts laterally (valgus)** pulled by **adductor hallucis** and the lateral pull of the **EHL and FHL** (**bowstring effect: become deforming forces**).
2. **The metatarsal head drifts medially off the sesamoids** (the **sesamoids stay in place: anchored by the adductor hallucis and the transverse metatarsal ligament**): **apparent lateral sesamoid displacement** relative to the metatarsal head; **crista erosion**.
3. **Pronation (eversion) of the hallux** (abductor hallucis migrates plantarly: loses abduction power; becomes a flexor/pronator).
4. **Lateral capsule and adductor contracture**; **transfer metatarsalgia** under the second metatarsal head (first ray loses load-bearing).
5. **Second toe deformities** (crossover toe, hammer toe), **second MTPJ instability/plantar plate tear**.

3. Evaluation

- **Weight-bearing AP and lateral radiographs** (and sesamoid axial view).
- **Normal values and severity:**

Angle	Normal	Mild	Moderate	Severe
Hallux valgus angle (HVA: first MT axis vs proximal phalanx axis)	**Under 15 degrees**	**15-20** (up to 20-25 in some schemes)	**20-40**	**Over 40**
Intermetatarsal angle (IMA: 1st-2nd MT)	**Under 9 degrees**	**9-11** (under 13)	**11-16** (13-20)	**Over 16** (over 20)
Distal metatarsal articular angle (DMAA)	**Under 10 degrees** (about 6-10)			
Hallux valgus interphalangeus (HVI)	**Under 10 degrees**			

- **Congruent vs incongruent joint: congruent (articular surfaces parallel; increased DMAA; juvenile)** vs **incongruent (subluxed: lateral subluxation of the phalanx on the head)**. **Congruent deformity: correction must not be done by soft-tissue release alone (would create an incongruent joint): use DMAA-correcting osteotomy (biplanar chevron) or extra-articular (double/triple) osteotomies**.
- **Sesamoid position (Hardy and Clapham: grades 1-7)**, **first MTPJ arthritis** (**hallux rigidus: fusion instead**), **first TMT hypermobility** (**dorsal translation over about 8-10 mm: Lapidus**), **metatarsus adductus** (**increased difficulty; IMA underestimated**), **pronation of the first metatarsal (round lateral head sign)**.

4. Management

Non-operative

- **Wide toe-box footwear, bunion pads, orthoses, toe spacers**; **do not correct the deformity**; **surgery is for pain, not cosmesis** ("**never operate for cosmesis**").

Operative principles: choose by severity, joint congruency, arthritis and TMT stability

Deformity	Procedure	Notes
Mild (HVA under 30, IMA under 13), congruent or incongruent	**Distal metatarsal osteotomy: Chevron (Austin)** (**V-shaped osteotomy at the metatarsal head, apex distal; lateral shift up to about 30-50% of the head width**) ± **distal soft-tissue procedure (lateral release)**	**AVN of the metatarsal head** (**lateral release + chevron: avoid excessive dorsolateral dissection: nutrient vessels enter dorsolaterally (first dorsal metatarsal artery) and plantar-laterally**); **biplanar chevron for increased DMAA**
Moderate (HVA 30-40, IMA 13-20)	**Scarf osteotomy** (**Z-shaped diaphyseal osteotomy**: lateral translation; **troughing (fragments telescope into each other: elevation of the head, loss of height)**), **proximal metatarsal osteotomy (proximal chevron, crescentic, opening/closing wedge, Ludloff, Mau)** + **distal soft-tissue release** (modified McBride)	**Proximal osteotomies: more powerful correction (each degree of correction at the base produces more displacement distally)**; **dorsal malunion (elevatus: transfer metatarsalgia)**
Severe (HVA over 40, IMA over 20)	**Proximal osteotomy + distal soft-tissue procedure**, **Lapidus (first TMT arthrodesis)**	

Deformity	Procedure	Notes
First TMT hypermobility / generalised laxity / recurrent HV	**Modified Lapidus (first TMT arthrodesis)**	**Non-union (about 5-10%)**, **shortening (transfer metatarsalgia)**, dorsiflexion malunion
Arthritis of the first MTPJ, severe deformity, neuromuscular, RA, salvage of failed surgery	**First MTPJ arthrodesis**	**Position: about 10-15 degrees valgus, about 10-15 degrees dorsiflexion relative to the floor (about 20-25 degrees relative to the first metatarsal shaft), neutral rotation**; **IMA corrects spontaneously** after fusion
Hallux valgus interphalangeus	**Akin osteotomy** (**medial closing-wedge osteotomy of the proximal phalanx**)	Adjunct to other osteotomies
Elderly, low-demand	**Keller resection arthroplasty** (**excision of the base of the proximal phalanx**)	**Cock-up deformity, transfer metatarsalgia, weakness of push-off**; largely abandoned
Juvenile/adolescent	**Delay until skeletal maturity** (**high recurrence**); **guided growth (lateral hemiepiphysiodesis of the first metatarsal)**	**Increased DMAA common**

- **Simple bunionectomy (Silver) alone**: **high recurrence; not used alone**.
- **McBride procedure** (original): **excision of the lateral sesamoid** (**hallux varus risk**), **adductor tenotomy**, **medial capsulorrhaphy**: **modified McBride = lateral release without sesamoid excision**.

- **Minimally invasive (MIS) third/fourth generation (MICA/percutaneous chevron-Akin)**: increasingly used; similar outcomes; learning curve.

5. Complications

- **Recurrence (commonest; about 10-25%)**: **undercorrection of the IMA**, **unrecognised increased DMAA**, **unaddressed pronation**, **first TMT hypermobility**, **juvenile**, **poor sesamoid reduction**.
- **Hallux varus** (**overcorrection: excessive lateral release, lateral sesamoid excision, excessive medial eminence resection (staking the sagittal sulcus), overtightened medial capsule, IMA overcorrection (negative)**): **flexible: EHB tenodesis/EHL transfer; rigid/arthritic: MTPJ fusion**.
- **AVN of the metatarsal head** (**chevron + extensive lateral release**).
- **Transfer metatarsalgia** (**shortening or dorsal malunion of the first metatarsal**).
- **Stiffness of the first MTPJ**, **non-union/malunion**, **infection**, **dorsomedial cutaneous nerve injury (neuroma)** (**medial dorsal cutaneous branch of the superficial peroneal nerve**), **CRPS**.

Part B: Hallux rigidus

6. Hallux rigidus (first MTPJ OA)

- **Commonest arthritis of the foot**; **dorsal osteophyte (dorsal bunion)** limiting **dorsiflexion (pain at push-off)**; **positive grind test (late)**; **adolescent form (OCD of the metatarsal head)**.
- **Coughlin and Shurnas classification (grades 0-4): 0 stiffness, normal radiograph**; **1 mild, dorsal osteophyte, preserved joint space**; **2 moderate, joint space narrowing, pain at extremes**; **3 severe, under 10-20% residual motion, pain at extremes, more narrowing**; **4: as 3 with pain in mid-range (positive grind test)**.
- **Treatment:**

- **Non-operative: stiff-soled shoe, rocker sole, Morton's extension (carbon plate)**, NSAIDs, steroid injection.
- **Cheilectomy** (**excision of the dorsal osteophyte and dorsal 25-30% of the metatarsal head**): **grades 1-2 (and selected 3)**; **pain at the extremes, not mid-range**; **± Moberg osteotomy (dorsal closing-wedge osteotomy of the proximal phalanx: shifts the arc into dorsiflexion)**.
- **First MTPJ arthrodesis**: **gold standard for grades 3-4** (**predictable pain relief**; **position: 10-15 degrees dorsiflexion relative to the floor, 10-15 degrees valgus, neutral rotation**).
- **Interpositional arthroplasty/Keller** (elderly), **hemiarthroplasty (synthetic cartilage implant, Cartiva: mixed results)**; **silicone/total joint arthroplasty: high failure (silicone synovitis, subsidence)**.

Part C: Lesser toe deformities and forefoot pain

7. Lesser toe deformities

Deformity	MTPJ	PIPJ	DIPJ	Notes
Mallet toe	Neutral	Neutral	**Flexed**	**FDL contracture**
Hammer toe	**Neutral or slightly extended**	**Flexed**	**Neutral/extended**	**Commonest; second toe**; FDL/FDB contracture
Claw toe	**Hyperextended**	**Flexed**	**Flexed**	**Usually multiple toes, bilateral: neuromuscular (CMT, diabetes neuropathy), RA, cavus foot: intrinsic weakness** (lumbrical/interossei)

Deformity	MTPJ	PIPJ	DIPJ	Notes
Curly toe		Flexed + varus (rotated)		**Congenital** (FDL tightness)

- **Flexible deformities: flexor-to-extensor transfer (Girdlestone-Taylor: FDL split to the extensor hood/dorsum of the proximal phalanx)**, **FDL tenotomy (mallet)**.
- **Fixed deformities: PIPJ resection arthroplasty (DuVries) or PIPJ arthrodesis**; **MTPJ: extensor tendon lengthening (EDL Z-lengthening), EDB tenotomy, dorsal capsulotomy**; **Weil osteotomy** (**oblique shortening osteotomy of the metatarsal neck**: for **MTPJ dislocation/subluxation, long metatarsal**: **floating toe** = commonest complication).
- **Crossover second toe / second MTPJ instability (plantar plate tear): dorsomedial drift of the second toe**, **pain under the second MTPJ**, **positive drawer (Lachman) test of the MTPJ**; **Coughlin plantar plate grading**; **plantar plate repair + Weil osteotomy**, flexor-to-extensor transfer.
- **Freiberg disease** (second metatarsal head osteochondrosis: adolescent girls; dorsal closing-wedge osteotomy).

8. Metatarsalgia

- **Pain under the metatarsal heads**: **primary** (**long second metatarsal**, plantarflexed metatarsal, **cavus foot**, prominent condyles), **secondary** (**inflammatory arthritis, Freiberg, MTPJ synovitis, gastrocnemius tightness**), **iatrogenic (transfer after hallux valgus surgery)**.
- **Intractable plantar keratosis (IPK)**: **discrete** (beneath a prominent fibular condyle) vs **diffuse** (beneath a long metatarsal).
- **Gastrocnemius contracture (positive Silfverskiöld test: ankle dorsiflexion limited with the knee extended, improves with the knee flexed)**: **gastrocnemius recession (Strayer)**.
- **Treatment: metatarsal pad (placed proximal to the metatarsal heads)**, rocker-bottom sole, orthoses; **Weil osteotomy**, **condylectomy**.

9. Morton neuroma (interdigital neuroma)

- **Perineural fibrosis of the common digital nerve** (**not a true neuroma**), **distal to the deep transverse metatarsal ligament**; **third web space (between 3rd and 4th metatarsal heads) commonest**, then second; **women (about 5-10:1)**, **narrow/high-heeled shoes**.
- **Burning plantar forefoot pain radiating into the toes**, **numbness**, **"walking on a pebble"**; **Mulder click (squeezing the forefoot mediolaterally while pressing the web space)**, **web space tenderness**; **splaying of the toes (Sullivan sign)**.
- **Imaging: ultrasound or MRI** (size **over about 5 mm**); **differentials: MTPJ synovitis/plantar plate tear (second web), stress fracture, intermetatarsal bursitis, tarsal tunnel syndrome, lumbar radiculopathy**.
- **Treatment: wide shoes, metatarsal pad, steroid injection (up to about 50-80% relief)**, **alcohol sclerosing injections**, **radiofrequency ablation**; **surgery: neurectomy (dorsal approach preferred: avoids plantar scar; divide the deep transverse metatarsal ligament; resect the nerve proximal to the ligament, about 3 cm)** (**about 80-90% success**); **recurrent (stump) neuroma (about 5-10%)**: **plantar approach** for revision; **decompression (ligament release) alone** as an alternative.

10. Sesamoid disorders

- **Sesamoiditis, stress fracture, AVN, osteochondritis, OA, bipartite sesamoid (medial/tibial sesamoid commonest: about 10-30%; bilateral in about 85%: compare with the other foot)**, **turf toe**.
- **Turf toe: hyperdorsiflexion injury of the first MTPJ (plantar plate/capsuloligamentous complex)**: **American football on artificial turf**; **grade I** sprain, **II** partial tear, **III** complete tear (**proximal migration of sesamoids**, **sesamoid diastasis**, **vertical instability: Lachman positive**); **grade III: surgical repair** of the plantar plate.
- **Treatment of sesamoid pain:** offloading (dancer's pad, orthoses), boot/cast; **sesamoidectomy** (refractory: **excision of both sesamoids**

causes cock-up deformity: avoid; **medial sesamoid excision: hallux valgus**; **lateral sesamoid excision: hallux varus**).

11. Bunionette (tailor's bunion)

- **Lateral prominence of the fifth metatarsal head**; **Coughlin types: I enlarged/prominent head**, **II lateral bowing of the fifth metatarsal**, **III increased 4-5 intermetatarsal angle (over 8-10 degrees: commonest symptomatic)**.
- **Treatment: wide shoes**; **type I: lateral condylectomy** or **distal chevron osteotomy**; **type II: distal chevron or diaphyseal (oblique/scarf) osteotomy**; **type III: diaphyseal oblique or proximal osteotomy** (**proximal fifth metatarsal osteotomy: poor blood supply: non-union**).

Exam pearl: **HVA under 15 and IMA under 9 normal; DMAA under 10. Sesamoids stay; the metatarsal head drifts medially. Congruent joint (increased DMAA, juvenile) needs DMAA-correcting osteotomy, not soft-tissue release. Mild: chevron; moderate: scarf/proximal osteotomy; severe/hypermobile: Lapidus; arthritic: fusion (10-15 valgus, 10-15 dorsiflexion to floor). Commonest complication: recurrence; hallux varus = overcorrection (lateral sesamoid excision).**

Exam pearl: **Hallux rigidus: Coughlin-Shurnas; cheilectomy for pain at the extremes (grade 1-2, ± Moberg); mid-range pain/grind (3-4) = fusion.**

Exam pearl: **Claw toe (MTPJ hyperextension): neuromuscular/cavus; flexible: Girdlestone-Taylor (FDL to extensor). Crossover second toe = plantar plate tear (drawer test): repair + Weil (floating toe). Morton neuroma: 3rd web, Mulder click, distal to DTML; neurectomy via dorsal approach. Bunionette type III (increased 4-5 IMA) = diaphyseal osteotomy.**

Achilles and tendons

Part A: Achilles tendon

1. Anatomy

- **Formed by the gastrocnemius (medial and lateral heads: cross the knee) and soleus** (± plantaris medially); **largest and strongest tendon in the body** (loads up to **about 12 times body weight** in running).
- **Fibres rotate about 90 degrees** (medial fibres become posterior, **soleus fibres insert medially**) as they descend: stress concentration.
- **Paratenon** (no true synovial sheath), **insertion on the middle third of the posterior calcaneal tuberosity** (footprint); **retrocalcaneal bursa** (between tendon and calcaneus) and **retro-Achilles (subcutaneous) bursa**.
- **Blood supply: posterior tibial artery (proximal and distal)**, **peroneal artery (mid-portion)**; **hypovascular watershed zone about 2-6 cm proximal to the insertion** (**site of most ruptures and mid-portion tendinopathy**).
- **Sural nerve**: crosses from medial to lateral, **lateral to the tendon at about 10 cm proximal to the insertion** (**at risk in lateral incisions and percutaneous repairs**).

2. Acute Achilles tendon rupture

- **Epidemiology: men aged 30-50 ("weekend warriors")**, **sudden eccentric loading** (push-off with knee extended, unexpected dorsiflexion), **left more than right**; **2-6 cm above the insertion (commonest)**.
- **Risk factors: fluoroquinolones**, **corticosteroids (systemic or local injection)**, **pre-existing tendinopathy**, inflammatory arthritis, gout, **renal failure, hyperparathyroidism**, previous contralateral rupture.
- **Clinical: "kicked in the back of the leg"**, **audible pop**, **weak plantarflexion (but often preserved through FHL, FDL, TP, peronei)**:

missed in about 20%.

- **Simmonds-Thompson test (calf squeeze: absent plantarflexion = rupture)** (**most reliable**),
- **Palpable gap**,
- **Matles test** (**loss of resting plantarflexion tension: the injured foot hangs in neutral/dorsiflexion with knees flexed prone**),
- **Increased passive dorsiflexion**, **O'Brien needle test**, **Copeland sphygmomanometer test**.
- **Two of the clinical tests (AAOS guideline) sufficient for diagnosis.**

- **Imaging: ultrasound** (gap, **gap distance in plantarflexion (over 5-10 mm)**: may guide treatment); **MRI** (chronic/equivocal); **radiograph: Kager's fat pad distortion (triangle obscured), avulsion fracture (calcaneal tuberosity avulsion: "beak" fracture: urgent surgery: skin necrosis)**.

Management

	Non-operative (functional rehabilitation)	Operative (open or minimally invasive)
Re-rupture	**About 3-12% (similar to surgery when early functional rehab is used)**	**About 2-5%**
Complications	Re-rupture; **tendon elongation (weaker push-off)**	**Wound complications/infection (about 5-10% open), sural nerve injury (percutaneous: about 5-10%)**, DVT
Strength	Possibly slightly weaker (elongation)	Slightly better plantarflexion strength in some studies

- **Evidence: early functional rehabilitation (early weight bearing in an equinus boot with heel wedges)** reduces the re-rupture difference: **Willits (2010)**, **UKSTAR trial (2020, UK: functional bracing vs plaster casting: similar outcomes)**, **Scandinavian trials (Nilsson-Helander,**

Ochen meta-analysis): **surgery reduces re-rupture absolute risk by about 2-3% at the cost of more complications**; **choice individualised** (**athletes, delayed presentation, large gap, avulsion: surgery**).

- **Non-operative protocol: equinus (about 20-30 degrees plantarflexion) in a boot with wedges**, **early protected weight bearing (2 weeks)**, **gradual reduction of equinus over 6-8 weeks**, **avoid passive dorsiflexion stretching for 12 weeks**; return to sport about 6-9 months.
- **Operative: open repair (Krackow/Bunnell core sutures + epitendinous; medial incision avoids the sural nerve)**, **minimally invasive/percutaneous (Achillon, PARS, Ma-Griffith)** (**fewer wound problems, sural nerve risk**); **repair with the ankle in resting tension (compare with the contralateral side)**.
- **Calcaneal tuberosity (sleeve) avulsion fracture: urgent ORIF (skin at risk)**.

3. Chronic (neglected) Achilles rupture

- **Over 4-6 weeks**; **gap with scar**; **weak push-off, inability to single heel rise**, **increased passive dorsiflexion**.
- **Kuwada classification** (gap): **I partial**, **II complete with gap under 3 cm (end-to-end repair)**, **III gap 3-6 cm (V-Y advancement / gastrocnemius recession / tendon graft)**, **IV gap over 6 cm (tendon transfer + graft)**.
- **Myerson: under 1-2 cm: end-to-end repair**; **2-5 cm: V-Y lengthening (gastrocnemius aponeurosis) ± FHL transfer**; **over 5 cm: FHL transfer (± turndown flap, allograft)**.
- **FHL transfer** (**strongest, in phase, adjacent, vascularised muscle belly brings blood supply, low donor morbidity**) to the calcaneus (**tunnel**); **FDL, peroneus brevis** alternatives; **gastrocnemius turndown flap (Bosworth/Lindholm)**.

4. Achilles tendinopathy

	Mid-portion (non-insertional)	Insertional
Location	**2-6 cm proximal to the insertion**	**At the calcaneal insertion**
Patients	**Runners, active middle-aged**	**Older, less active, overweight**, inflammatory arthritis (seronegative)
Pathology	**Tendinosis (degeneration, neovascularisation)**, **paratenonitis** (crepitus, fixed tender area)	**Calcification/enthesophyte (spur)**, **Haglund deformity (posterosuperior calcaneal prominence)**, **retrocalcaneal bursitis**
Signs	**Painful fusiform swelling that moves with the tendon (painful arc sign)**; **Royal London Hospital test**	**Posterior heel tenderness, bump ("pump bump")**
First-line	**Eccentric loading (Alfredson protocol: 3 x 15 reps, twice daily, 12 weeks, knee straight and bent)** or **heavy slow resistance**; **ESWT**; **high-volume injection**, sclerotherapy	**Eccentric loading on flat ground (not into dorsiflexion)**, **heel lift**, **shoe modification (open-backed)**, **ESWT**
Surgery (after 6 months)	**Debridement/longitudinal tenotomies**, **stripping of paratenon/neovessels**, **plantaris release (medial)**, **gastrocnemius recession**, **FHL transfer if over 50% of tendon debrided**	**Haglund resection + debridement of calcification + reattachment (suture anchors)** (**central tendon-splitting approach**); **FHL transfer** for extensive degeneration (over 50%); **Zadek (dorsal closing-wedge calcaneal) osteotomy**

- **Haglund syndrome: Haglund deformity + retrocalcaneal bursitis + insertional tendinopathy**; **radiographic: Fowler-Philip angle (over 75 degrees)**, **parallel pitch lines (Pavlov)**.

- **Avoid steroid injections into the Achilles** (rupture); **fluoroquinolones** caution.

Part B: Other tendons

5. Peroneal tendons

- **Anatomy: peroneus brevis (anterior/medial, closer to the fibula, in the retromalleolar groove; inserts on the base of the fifth metatarsal)**, **peroneus longus (posterior/lateral; crosses under the cuboid in the cuboid tunnel to insert on the base of the first metatarsal and medial cuneiform)**; **superior peroneal retinaculum (SPR: from the posterolateral fibula to the calcaneus)**; **fibrocartilaginous rim** deepens the groove; **os peroneum** (sesamoid in PL at the cuboid: about 20%).
- **Peroneus brevis split tears: longitudinal splits from compression against the fibula** (**cavovarus foot, ankle instability, low-lying PB muscle belly, peroneus quartus**): **lateral ankle pain/swelling**; **MRI ("boomerang/chevron" shape)**; **debridement and tubularisation (if under 50% involved)**, **tenodesis to PL (if over 50%)**; **correct cavovarus**.
- **Peroneal tendon subluxation/dislocation: forceful dorsiflexion with peroneal contraction (skiing)**; **SPR avulsion with a "fleck sign" (rim fracture of the lateral malleolus: Eckert-Davis grade 3)**; **Eckert-Davis/Oden grades (1: SPR stripped from the fibula with periosteum (commonest); 2: fibrocartilaginous rim elevated with SPR; 3: bony avulsion; 4: SPR torn from calcaneal attachment)**. **Acute: cast trial (about 50% success) or SPR repair**; **chronic: SPR repair/reconstruction + fibular groove deepening**. **Intrasheath subluxation** (tendons swap position within the groove).
- **Painful os peroneum syndrome (POPS): PL rupture at the cuboid notch/os peroneum fracture**; **proximal migration of the os peroneum on serial radiographs = PL rupture**; excision + PB tenodesis.

6. Tibialis posterior

- See flatfoot notes (PCFD).

7. Flexor hallucis longus

- **FHL tenosynovitis ("dancer's tendinitis")**: **ballet dancers (en pointe)**: **posteromedial ankle pain, triggering of the hallux ("hallux saltans")**, **pain on resisted hallux flexion**; **stenosis at the fibro-osseous tunnel behind the talus (between the medial and lateral talar tubercles)**; associated with **posterior ankle impingement (os trigonum/Stieda process)**: **rest, physiotherapy; release of the FHL tunnel ± os trigonum excision (open posteromedial or arthroscopic/endoscopic posterior)**.
- **Knot of Henry**: FHL-FDL crossover (plantar midfoot).

8. Tibialis anterior

- **Rupture: elderly men, degenerative**; see flatfoot notes.

9. Posterior ankle impingement

- **Os trigonum** (**unfused lateral tubercle of the posterior talar process: about 7-14%**) or **Stieda process** (elongated lateral tubercle); **ballet dancers, footballers (forced plantarflexion: nutcracker)**; **posterior ankle pain in plantarflexion (forced plantarflexion test)**, **diagnostic injection**; **excision (posterior arthroscopy: posterolateral and posteromedial portals with the patient prone; FHL is the landmark: stay lateral to FHL to protect the tibial neurovascular bundle)**.
- **Anterior ankle impingement: "footballer's ankle"**: **anterior tibial and talar neck osteophytes (kissing)** (**bony**) or **anterolateral soft-tissue impingement (Bassett ligament: accessory anteroinferior tibiofibular ligament fascicle; meniscoid lesion (Wolin))**; **arthroscopic debridement**; **anterior arthroscopy portals: anteromedial (medial to the tibialis anterior) and anterolateral (lateral to peroneus tertius: superficial peroneal nerve (intermediate dorsal cutaneous branch)**

at risk: commonest complication of ankle arthroscopy); **avoid the anterocentral portal (dorsalis pedis/deep peroneal nerve)**.

10. Plantar fascia rupture

- **Acute rupture** (often after steroid injection), **plantar heel pain and bruising**; non-operative (boot).

11. Gastrocnemius-soleus problems

- **Tennis leg** (**medial head of gastrocnemius tear at the musculotendinous junction**; middle-aged; "shot in the calf"); **differentiate from DVT (ultrasound)**; non-operative.
- **Isolated gastrocnemius contracture** (**Silfverskiöld test**): associated with **metatarsalgia, plantar fasciitis, Achilles tendinopathy, PCFD, midfoot arthritis**: **gastrocnemius recession (Strayer: distal aponeurosis; Baumann: intramuscular proximal)** (**sural nerve at risk**); **Achilles lengthening (Hoke triple hemisection: percutaneous)** when both gastrocnemius and soleus are tight (**overlengthening: calcaneal gait/weak push-off**).

> Exam pearl: **Achilles: watershed 2-6 cm above insertion; sural nerve lateral at about 10 cm. Thompson test most reliable. Early functional rehab narrows re-rupture difference (UKSTAR, Willits); surgery: fewer re-ruptures, more wound/sural complications. Chronic gap over 5 cm: FHL transfer (± V-Y/turndown).**

> Exam pearl: **Mid-portion tendinopathy: eccentric Alfredson (12 weeks), plantaris release. Insertional: no dorsiflexion eccentrics; Haglund resection + debridement + reattachment; FHL if over 50% debrided. Avoid steroid injections and fluoroquinolones.**

Exam pearl: **Peroneus brevis split tears (cavovarus): under 50% repair, over 50% tenodesis to PL. Peroneal dislocation: SPR, fleck sign; groove deepening for chronic. Os peroneum migration = PL rupture. Dancers: FHL tendinitis + os trigonum (stay lateral to FHL arthroscopically). Ankle arthroscopy: anterolateral portal risks the superficial peroneal nerve.**

Ankle arthritis and instability

Part A: Ankle instability

1. Lateral ligament complex

- **Anterior talofibular ligament (ATFL): weakest and most commonly injured**; **intracapsular**; **taut in plantarflexion** (resists **anterior translation and inversion in plantarflexion**); from the anterior fibula about **1 cm proximal to the tip** to the talar neck/body.
- **Calcaneofibular ligament (CFL): extracapsular**, **deep to the peroneal tendons (floor of the peroneal sheath)**; **taut in dorsiflexion**; resists **inversion in neutral/dorsiflexion**; **crosses both the ankle and subtalar joints**.
- **Posterior talofibular ligament (PTFL):** strongest; **rarely injured** (only in dislocations).
- **Injury sequence (inversion in plantarflexion): ATFL first, then CFL, then PTFL.**

2. Acute lateral ankle sprain

- **Commonest sports injury**; **inversion + plantarflexion**.
- **Grading: I** stretch, **II** partial tear (ATFL torn), **III** complete tear (ATFL + CFL).
- **Ottawa ankle rules (radiograph if): pain in the malleolar zone AND** (**bone tenderness at the posterior edge or tip of either malleolus (distal 6 cm)**, or **inability to bear weight (4 steps) immediately and in the emergency department**); **foot: navicular or base of 5th metatarsal tenderness** (**sensitivity close to 100%**; reduce radiographs by about 30-40%).
- **Examination (delayed 4-5 days improves accuracy): anterior drawer test** (**ATFL**; **over 10 mm absolute or over 3-5 mm side-to-side difference = abnormal**), **talar tilt test** (**CFL**; **over 10-15 degrees or over 5 degrees side-to-side**).

- **Differential diagnosis/associated injuries: high ankle (syndesmotic) sprain**, **osteochondral lesion of the talus (OLT)**, **peroneal tendon injury**, **fractures: lateral process of the talus ("snowboarder's fracture"), anterior process of the calcaneus, base of the 5th metatarsal, os trigonum**, **Maisonneuve**, **deltoid injury**, **subtalar instability**.
- **Treatment: functional treatment** (**brief protection (brace/boot for severe), early mobilisation, proprioceptive and peroneal strengthening**) **is superior to immobilisation** for most; **grade III: short below-knee cast/boot (about 10 days) improves early outcomes (Lamb, CAST trial)**; **acute surgical repair rarely indicated** (elite athletes debatable).
- **Chronic symptoms after sprains in about 20-40%** (pain, instability).

3. Chronic lateral ankle instability

- **Mechanical instability** (**pathological laxity: positive drawer/tilt**) vs **functional instability** (**subjective giving way: proprioceptive deficit, peroneal weakness**).
- **Evaluate: hindfoot alignment (cavovarus: Coleman block test)**, **generalised laxity (Beighton)**, **peroneal tendons (split tears)**, **OLT**, **anterior impingement**, **subtalar instability**, **syndesmosis**.
- **Imaging: stress radiographs** (**anterior drawer: over 10 mm absolute or 3 mm side-to-side; talar tilt over 10 degrees or 5 degrees side-to-side**), **MRI** (associated lesions), **ultrasound**.
- **Non-operative first: peroneal strengthening and proprioceptive (balance board) rehabilitation for at least 3-6 months**, braces.
- **Surgery:**
  - **Anatomic repair: modified Broström-Gould procedure (gold standard)** (**imbrication/shortening of the ATFL and CFL + reinforcement with the inferior extensor retinaculum (Gould modification)**): **about 85-95% good results**; **arthroscopic Broström** increasingly used.

- **Augmentation (suture tape/internal brace)** for poor tissue, hyperlaxity, high-demand.
- **Anatomic reconstruction with free tendon graft (gracilis/semitendinosus/allograft)**: **failed Broström, poor-quality tissue, generalised laxity (Ehlers-Danlos), heavy patients, long-standing instability**.
- **Non-anatomic tenodeses (peroneus brevis): Watson-Jones, Evans, Chrisman-Snook**: **historical**: **restrict subtalar motion, stiffness, sacrifice of PB**.
- **Correct cavovarus (lateralising calcaneal osteotomy, first metatarsal dorsiflexion osteotomy)** if present: **otherwise failure**.
- **Concomitant ankle arthroscopy** for intra-articular lesions (OLT, impingement).

- **Complications: superficial peroneal nerve injury (commonest)**, **sural nerve injury**, **stiffness (loss of inversion)**, recurrence.

4. Syndesmotic (high ankle) sprains

- **External rotation + dorsiflexion**: **AITFL (first), interosseous ligament, PITFL, transverse ligament**; **collision sports**; **longer recovery (about 2x lateral sprains)**.
- **Tests: squeeze test (Hopkin)**, **external rotation stress test (Kleiger: most sensitive)**, **crossed-leg test**, **Cotton test (lateral translation of the talus)**, **fibular translation test**.
- **Radiographs (weight-bearing/stress): tibiofibular clear space over 6 mm (1 cm above the plafond on AP and mortise: most reliable)**, **tibiofibular overlap under 6 mm on AP (or under 1 mm on mortise)**, **medial clear space over 4-5 mm (deltoid injury)**; **bilateral weight-bearing CT** (subtle diastasis), **MRI**; **arthroscopy (gold standard: over 2 mm diastasis on probing)**.
- **Treatment: stable (no diastasis): non-operative (boot, protected weight bearing)**; **unstable/diastasis: fixation (syndesmotic screws (3.5-4.5 mm, 1-2 screws, 3-4 cortices, about 2-5 cm above the plafond, angled 20-30 degrees anteriorly from posterior) or suture-button**

devices (dynamic; fewer removals, better reduction)); **anatomic reduction of the fibula in the incisura is the key (malreduction = poor outcome)**; **direct visualisation/CT** recommended.

- **Chronic syndesmotic instability:** debridement + fixation/reconstruction; **arthrodesis** of the syndesmosis (salvage).

5. Medial (deltoid) ligament injury and instability

- **Superficial deltoid** (tibiospring, tibionavicular, tibiocalcaneal: **resists hindfoot eversion**) and **deep deltoid (anterior and posterior tibiotalar: primary restraint to lateral talar shift and external rotation)**.
- **Isolated deltoid injuries are rare** (usually with **fractures (bimalleolar equivalent)** or syndesmotic injury); **chronic medial instability** associated with **PCFD (stage IV)** and valgus ankles.
- **Gravity stress test / external rotation stress test**: **medial clear space over 4-5 mm** (or **over 1 mm greater than the superior clear space**) = deep deltoid insufficiency.
- **Repair/reconstruction (anatomic, anchors, graft)**.

6. Subtalar instability

- Often **coexists with lateral ankle instability (CFL, cervical and interosseous talocalcaneal ligaments)**; difficult to diagnose (**Brodén stress views**); **reconstruct the CFL** (anatomic reconstruction addressing both joints).

Part B: Ankle arthritis

7. Epidemiology and aetiology

- **Post-traumatic (about 70-80%: rotational ankle fractures, pilon fractures, recurrent instability)**, **primary OA (uncommon: about 10%)**, **inflammatory (RA)**, **haemophilia**, **neuropathic (Charcot)**, **AVN of the talus**, **septic sequelae**, **varus/valgus malalignment (cavovarus, PCFD)**.

- **Ankle cartilage is thinner (about 1-1.7 mm) but more resistant to OA** than the knee (**higher proteoglycan content, stiffer, less response to catabolic cytokines (IL-1), smaller contact area but high congruency**): **younger patients (post-traumatic) than hip/knee OA**; **disability comparable to end-stage hip OA or heart failure (SF-36)**.

8. Evaluation

- **Weight-bearing AP, mortise and lateral ankle + foot radiographs**, **hindfoot alignment view (Saltzman)**, **long-leg alignment**; **weight-bearing CT**.
- **Classifications: Takakura (varus OA: stage 1 sclerosis/osteophytes, 2 medial narrowing, 3A medial gutter obliteration, 3B narrowing to the talar dome, 4 whole joint)**; **Kellgren-Lawrence**; **COFAS** (for TAR planning).
- **Assess: coronal deformity (varus/valgus: congruent vs incongruent)**, **ligament stability**, **hindfoot/subtalar arthritis**, **bone stock**, **soft tissues**, **vascular status**, **neuropathy**, **adjacent joint arthritis**, **smoking**, **infection history**.

9. Non-operative treatment

- **Activity modification, weight loss, NSAIDs, bracing (lace-up brace, AFO (Arizona)), rocker-bottom sole with SACH heel (solid ankle cushioned heel)**, **intra-articular steroid/hyaluronic acid (limited evidence)**, PRP (experimental).

10. Joint-preserving surgery

- **Arthroscopic debridement (anterior impingement osteophytes)**: **for impingement symptoms with preserved joint space**; poor for diffuse arthritis.
- **Supramalleolar osteotomy (SMOT):**
  - **Indication: asymmetric (varus or valgus) ankle OA with partially preserved (at least 50%) joint surface** (**Takakura 2-3A**), **younger patients**.

  - **Medial opening-wedge (or lateral closing-wedge) tibial osteotomy for varus OA** (**overcorrect slightly to about 2-4 degrees valgus**) ± **fibular osteotomy**; **medial closing-wedge for valgus OA**.
  - **Contraindications: end-stage panarthritis**, **severe hindfoot instability**, **neuropathy**, **inflammatory arthritis**, **severe talar tilt (incongruent joint with over about 10 degrees tilt: relative)**.
  - **Delays fusion/TAR** (about 10 years in many patients).
- **Distraction arthroplasty (circular frame distraction 5 mm for about 3 months)**: **young patients**; cartilage repair; variable results.
- **Osteochondral allograft (total ankle allograft)**: high failure; limited role.

11. Ankle arthrodesis

- **Gold standard for end-stage ankle arthritis** (especially **young, heavy laborers, infection history, neuropathy, severe deformity, poor bone stock (talus), failed TAR**).
- **Position: neutral dorsiflexion (90 degrees), 5 degrees hindfoot valgus, external rotation about 5-10 degrees (equal to the contralateral foot), talus translated slightly posteriorly (decreases lever arm/knee hyperextension)**; **equinus position = genu recurvatum/knee pain; varus = lateral foot overload**.
- **Techniques: arthroscopic arthrodesis** (**faster union (about 8-12 weeks), lower complications, less blood loss** in mild deformity: **under about 10-15 degrees**), **open (anterior or transfibular lateral approach)** for deformity/bone loss; **fixation: 2-3 crossed cannulated cancellous screws (6.5-7.3 mm)** ("**home-run screw**" from the posterior malleolus into the talar neck), **anterior plates**, **TTC nail (when subtalar also fused)**.
- **Union rate about 85-95%**; **non-union about 5-15%** (**smoking (strongest modifiable risk), diabetes, AVN of the talus, open approach, infection, previous fracture**).
- **Long-term: adjacent joint arthritis (subtalar about 50-100% radiographically, talonavicular, midfoot) at about 7-20 years**; gait:

reduced sagittal motion compensated by midfoot (about 30% of ankle motion remains through adjacent joints); **shoe wear: rocker sole**.

- **Fusion takedown/conversion to TAR** for painful adjacent joint disease (selected).

12. Total ankle replacement (TAR)

- **Modern designs: 2-component (fixed-bearing: INBONE, Infinity, STAR (US: some versions), Salto Talaris)** and **3-component (mobile-bearing: STAR (Europe), Hintegra, Mobility (withdrawn), BOX)**.
- **Indications: older (over about 50-55), lower-demand patients with end-stage OA**, **bilateral ankle arthritis**, **adjacent hindfoot/midfoot arthritis or fusion (TAR preserves motion: protects adjacent joints)**, **inflammatory arthritis**, **good bone stock and soft tissues**, **correctable deformity (under about 10-15 degrees coronal; up to 20 degrees with adjunctive procedures)**.
- **Contraindications: active or recent infection**, **Charcot neuroarthropathy/neuropathy**, **AVN of the talus (large)**, **severe PAD**, **severe uncorrectable deformity/instability**, **poor soft-tissue envelope**, **young heavy labourers (relative)**, **non-reconstructable ligaments**, **paralysis**.
- **Outcomes: survivorship about 70-90% at 10 years** (registry: **revision rates higher than hip/knee arthroplasty**); **functional outcomes comparable to arthrodesis (TARVA trial, UK 2022: similar AOFAS/MOXFQ at 1 year; TAR slightly better in some subgroups)**; **better gait, preserved motion, less adjacent joint degeneration**, but **more reoperations**.
- **Complications: wound complications (commonest; anterior approach: about 5-15%; thin skin)**, **intraoperative malleolar fracture (medial malleolus)**, **aseptic loosening/subsidence (talar component commonest)**, **periprosthetic osteolysis/cysts**, **gutter impingement (medial/lateral; heterotopic ossification)**, **polyethylene wear/fracture**, **infection**, **nerve injury (superficial and deep peroneal)**, **edge loading (malalignment)**.

- **Failed TAR: revision TAR (custom/stemmed talar components)** or **conversion to arthrodesis (TTC nail with bulk allograft (femoral head): "salvage arthrodesis")**: **union rates lower (about 80%)**.
- **Adjunctive procedures to balance the TAR: deltoid release (varus)**, **lateral ligament reconstruction**, **calcaneal osteotomy**, **subtalar/TN fusion**, **gastrocnemius recession**, **SMOT** (staged).

13. Osteochondral lesions of the talus (OLT) (link)

- **Medial (posteromedial: deeper, cup-shaped, often atraumatic: commonest location)** and **lateral (anterolateral: shallow, wafer-like, traumatic)**.
- **Berndt and Harty (radiographic: I compression, II partial detachment, III complete detachment, non-displaced, IV displaced fragment)**, **Hepple (MRI)**, **Ferkel (CT)**.
- **Treatment: non-operative (3-6 months) for non-displaced**; **arthroscopic debridement + bone marrow stimulation (microfracture)** for **lesions under about 1-1.5 cm² (or under 15 mm diameter)**; **fixation of large acute fragments**; **osteochondral autograft (OATS from the knee) or allograft**, **ACI/MACI** for **large (over 1.5 cm²) or cystic lesions or failed microfracture**; **medial malleolar osteotomy** for posteromedial access (**malunion/non-union risk**).

> Exam pearl: **ATFL weakest, taut in plantarflexion, intracapsular, injured first; CFL extracapsular, under the peroneals, taut in dorsiflexion, crosses ankle and subtalar joints. Ottawa rules: malleolar bone tenderness at posterior edge/tip of distal 6 cm or can't bear weight 4 steps.**

> Exam pearl: **Chronic instability: rehab 3-6 months first; modified Broström-Gould (inferior extensor retinaculum) gold standard; correct cavovarus (else failure); superficial peroneal nerve commonest nerve injury. Syndesmosis: clear space over 6 mm (most reliable), ER stress test most sensitive; anatomic fibular reduction; suture button vs screw.**

Exam pearl: **Ankle OA mostly post-traumatic. Asymmetric OA with over 50% joint preserved = SMOT. Arthrodesis position: neutral, 5 valgus, 5-10 ER, posterior talus; smoking = non-union; adjacent subtalar arthritis long-term. TAR: older, bilateral, adjacent fusion, inflammatory; contraindicated in infection, Charcot, talar AVN; wound problems commonest; TARVA similar to fusion.**

Diabetic foot and Charcot

Part A: The diabetic foot

1. Epidemiology and pathogenesis

- **Lifetime risk of foot ulceration in diabetes: about 15-25%**; **about 85% of diabetic lower-limb amputations are preceded by an ulcer**; **5-year mortality after major amputation is about 50-70%** (worse than many cancers). **India has one of the largest diabetic populations** (barefoot walking, late presentation, infection).
- **Triad:**
  - **Peripheral neuropathy (commonest underlying cause of ulcers; about 80-90%)**:
  - **Sensory** (loss of protective sensation: **10-g Semmes-Weinstein monofilament**: **inability to feel it = loss of protective sensation**; **128 Hz tuning fork (vibration)**, **ankle reflexes**, **biothesiometer (VPT over 25 V)**),
  - **Motor** (intrinsic weakness: **claw toes, prominent metatarsal heads, fat pad displacement**: high plantar pressures),
  - **Autonomic** (**dry, cracked skin (anhidrosis)**, **arteriovenous shunting: warm foot with bounding pulses**, increased bone blood flow: osteopenia).
  - **Peripheral arterial disease (PAD)**: **infrapopliteal (tibial/peroneal) arteries**, **foot arteries often spared**; **medial arterial calcification (Mönckeberg sclerosis) makes ABI falsely high (over 1.3: non-compressible)**: **use toe pressures (over 30-40 mmHg needed for healing), toe-brachial index (TBI under 0.7 abnormal), transcutaneous oxygen pressure (TcPO2 over 30-40 mmHg predicts healing)**, **arterial Doppler waveforms**, **angiography**.
  - **Infection** (impaired immunity: neutrophil dysfunction with hyperglycaemia).

- **Other factors: deformity** (hallux valgus, claw toes, Charcot), **limited joint mobility (glycosylation of collagen: stiff joints, Achilles contracture: forefoot pressure)**, **previous ulcer/amputation (strongest predictor of new ulcer)**, poor glycaemic control, **renal failure (dialysis)**, smoking, poor footwear, visual impairment.

2. Classification of ulcers

Wagner (Meggitt-Wagner) classification

Grade	Description
0	Intact skin (pre-ulcerative lesion, deformity, healed ulcer)
1	Superficial ulcer (skin and subcutaneous)
2	Deep ulcer to tendon, capsule or bone (no abscess/osteomyelitis)
3	Deep ulcer with abscess, osteomyelitis or septic arthritis
4	Localised (forefoot/toe) gangrene
5	Whole-foot gangrene

- **Limitation: does not account for ischaemia or infection separately**.

University of Texas (UT) classification

- **Grade (depth): 0 pre/post-ulcerative, 1 superficial, 2 to tendon/capsule, 3 to bone/joint**, **Stage: A clean, B infected, C ischaemic, D infected and ischaemic** (**stage predicts amputation better than grade**).

Other

- **SINBAD** (Site, Ischaemia, Neuropathy, Bacterial infection, Area, Depth: **IWGDF recommended for communication**), **WIfI** (**Wound, Ischaemia, foot Infection**: **SVS, predicts amputation risk and benefit of revascularisation**), **PEDIS** (IWGDF infection classification: perfusion, extent, depth, infection, sensation).
- **IWGDF/IDSA infection severity: uninfected**, **mild (local, cellulitis under 2 cm)**, **moderate (cellulitis over 2 cm or deeper tissue: abscess,**

osteomyelitis, septic arthritis, fasciitis), **severe (SIRS/systemic toxicity)**.

3. Assessment of the ulcer

- **Probe-to-bone test** (**sterile metal probe touching bone**): **high positive predictive value for osteomyelitis in high-prevalence settings** (and high negative predictive value in low-prevalence settings).
- **Radiographs** (**osteomyelitis changes appear after 10-14 days: periosteal reaction, cortical erosion, osteolysis**; gas in soft tissue; foreign bodies), **MRI (most sensitive and specific imaging for osteomyelitis: marrow oedema on T1 hypointense (confluent low T1 signal is the most specific), T2/STIR hyperintense; sinus tracts, abscesses)**, **labelled leucocyte scan/PET-CT** (when MRI contraindicated; helps differentiate Charcot from osteomyelitis).
- **Bone biopsy (histology + culture): gold standard for osteomyelitis diagnosis** (percutaneous through intact skin, or at surgery); **superficial swab cultures are unreliable** (deep tissue samples after debridement).
- **Inflammatory markers: ESR over 70 mm/h suggests osteomyelitis**; CRP, procalcitonin.
- **Organisms:**
  - **Mild/acute infections: aerobic gram-positive cocci (*S. aureus*, beta-haemolytic streptococci)**,
  - **Chronic/previously treated/severe: polymicrobial (gram-negatives: *Pseudomonas* (macerated wounds), Enterobacteriaceae; anaerobes (necrosis/ischaemia)), MRSA**.

4. Management principles

- **Multidisciplinary diabetic foot team** (**reduces major amputation rates by about 40-80%**): diabetologist, orthopaedic/vascular surgeon, podiatrist, orthotist, wound nurse, microbiologist.
- **Offloading (the cornerstone for neuropathic plantar ulcers):**
  - **Total contact cast (TCC): gold standard** (**forces compliance: non-removable**; **contraindicated with infection, ischaemia,**

deep/draining wounds; change weekly; **healing about 80-90% in 6-8 weeks**),
    - **Non-removable knee-high walker ("instant TCC")** (**equivalent**), removable walkers (compliance issues), **felted foam**, therapeutic footwear.
- **Debridement** (**sharp debridement of callus and necrotic tissue: converts a chronic wound to an acute one**), **moist wound healing**, **negative pressure wound therapy (NPWT: after debridement, post-amputation wounds)**, adjuncts (**hyperbaric oxygen (selected ischaemic wounds)**, **growth factors (becaplermin: PDGF)**, skin substitutes, **sucrose octasulfate dressings**).
- **Revascularisation** (**endovascular angioplasty or bypass**) for PAD before definitive surgery.
- **Infection:**
    - **Mild: oral antibiotics (covering gram-positive cocci) 1-2 weeks.**
    - **Moderate/severe: urgent surgical drainage/debridement (abscess, necrotising infection, gas, compartment involvement) + IV broad-spectrum antibiotics**; **"diabetic foot attack" = emergency**.
    - **Osteomyelitis: surgical resection of infected bone** (**± 1-3 weeks antibiotics after complete resection with clean margins**) or **prolonged antibiotics alone (6 weeks) in selected forefoot cases**.
- **Glycaemic control, nutrition, smoking cessation, renal optimisation.**

5. Surgical procedures

- **Prophylactic/corrective ("elective, curative, emergent" classification):**
    - **Achilles tendon lengthening (percutaneous Hoke)/gastrocnemius recession**: **forefoot plantar ulcers** (**reduces forefoot pressure: lowers recurrence (about 75% reduction at 2 years, Mueller)**: **transfer ulcers to the heel (overlengthening)**),
    - **Flexor tenotomy (percutaneous)** for **apical/tip ulcers of claw toes**,
    - **Metatarsal head resection (single or panmetatarsal)**, **sesamoidectomy**, **exostectomy** (Charcot rocker-bottom

prominences), **hallux IPJ arthroplasty (Keller) for plantar hallux ulcers**.

- **Amputations (level by perfusion, function, soft-tissue coverage):**
  - **Toe/ray amputation**: **hallux amputation: transfer pressure to the lesser metatarsals**; **second toe amputation: hallux valgus**; **ray amputation (single lateral rays tolerated; first ray amputation: poor)**.
  - **Transmetatarsal amputation (TMA)**: good function; **balance with Achilles lengthening and tibialis anterior/peroneal considerations** (equinovarus deformity tendency).
  - **Lisfranc/Chopart amputations**: **equinus and varus deformity** (**tibialis anterior/posterior unopposed; Achilles unopposed**): **Achilles tenotomy + tibialis anterior transfer to the talar neck/lateral**.
  - **Syme amputation** (ankle disarticulation with **heel pad preserved and anchored to the tibia**; **heel pad migration**: commonest complication; end-bearing; needs a patent posterior tibial artery).
  - **Below-knee (transtibial) amputation (BKA)**: **preserve knee: energy expenditure about 10-40% above normal (BKA) vs about 60-100% (AKA)**; **long posterior myocutaneous flap (Burgess)**; **ideal tibial length about 12-15 cm below the joint line (about 2.5 cm per 30 cm of height)**; **fibula 1-2 cm shorter than the tibia**.
  - **Through-knee/above-knee**: non-ambulators, failed BKA, fixed knee contracture.
- **Minimum requirements for healing: ABI over about 0.45**, **toe pressures over about 30-40 mmHg**, **TcPO2 over 30-40 mmHg**, **serum albumin over 3.0 g/dL**, **total lymphocyte count over 1,500/µL**.

6. Prevention

- **Annual foot screening for all diabetics** (**monofilament, vibration, pulses, deformity, skin, footwear**); **risk stratification** (IWGDF 0-3).
- **Education (daily self-inspection, never barefoot, moisturise, nail care)**, **therapeutic footwear/insoles** (**extra-depth shoes, custom**

insoles: prevent recurrence), **podiatry (callus removal)**, **temperature monitoring (home: differences over 2.2 C between feet = impending ulcer or Charcot)**.

Part B: Charcot neuroarthropathy

7. Definition and pathogenesis

- **Progressive, non-infectious, destructive inflammatory process of bones and joints in a neuropathic foot**, leading to **fragmentation, dislocation, collapse and deformity**.
- **Causes: diabetes (commonest today, about 0.1-0.4% of diabetics; up to about 13% of high-risk)**, **leprosy (India)**, **syringomyelia (upper limb: shoulder/elbow)**, **tabes dorsalis (syphilis: knee, hip)** (historical commonest), **alcoholic neuropathy, spinal cord injury, congenital insensitivity to pain, myelomeningocele (ankle)**.
- **Theories:**
  - **Neurotraumatic (German) theory**: **repetitive unperceived microtrauma in an insensate foot**,
  - **Neurovascular (French) theory**: **autonomic neuropathy: hyperaemia (AV shunting), osteoclastic bone resorption, osteopenia**,
  - **Unifying modern theory: RANKL-OPG pathway**: **trauma triggers uncontrolled inflammation (TNF-alpha, IL-1beta, IL-6): increased RANKL, osteoclast activation**; **loss of protective pain response** allows continued loading.

8. Eichenholtz stages (natural history) (+ Shibata stage 0)

Stage	Name	Clinical	Radiographs
0 (Shibata, "prodromal/at-risk")	Inflammation	Warm, swollen, red foot after minor trauma	Normal (MRI: bone marrow oedema, microfractures)

Stage	Name	Clinical	Radiographs
I	Development (fragmentation)	Warm (over 2 C higher), swollen, erythematous; may be painless	Fragmentation, debris, joint subluxation/dislocation, osteopenia
II	Coalescence	Decreased warmth/swelling	Absorption of debris, sclerosis, early fusion of fragments
III	Reconstruction (consolidation)	No warmth; stable (possibly deformed) foot	Remodelling, bony ankylosis, deformity (rocker-bottom)

- **Each stage lasts months; the whole process about 1-2 years**.
- **Erythema that resolves on elevation (Brodsky elevation test: within about 1 minute)** suggests **Charcot rather than infection** (cellulitis does not resolve on elevation).

9. Anatomical classification (Brodsky; modified)

Type	Region	Frequency	Notes
1	Midfoot: tarsometatarsal (Lisfranc) and naviculocuneiform joints	Commonest (about 60-70%)	Rocker-bottom deformity with plantar ulceration (cuboid or medial cuneiform prominence); relatively stable
2	Hindfoot: subtalar, talonavicular, calcaneocuboid (Chopart)	About 10-30%	Instability; long time to consolidate
3A	Ankle (tibiotalar)	About 10%	Most unstable; varus/valgus; ulceration over malleoli; worst prognosis

Type	Region	Frequency	Notes
3B	**Calcaneal tuberosity (pathological avulsion fracture)**	Rare	**Loss of push-off, flatfoot**
(**4**)	**Multiple regions**		
(**5**)	**Forefoot (MTPJs)**		

- **Sanders and Frykberg classification (patterns I-V): I forefoot (MTPJ/IPJ)**, **II tarsometatarsal (Lisfranc)**, **III naviculocuneiform/TN/CC (Chopart)**, **IV ankle/subtalar**, **V calcaneus**.
- **Rogers and Bevilacqua classification: anatomical location (1 forefoot, 2 midfoot, 3 hindfoot/ankle)** + **A no deformity, B deformity, C ulcer, D osteomyelitis**.

10. Diagnosis

- **Hot, swollen foot in a neuropathic diabetic: Charcot until proved otherwise** (often **misdiagnosed as cellulitis, gout, DVT or sprain**).
- **Temperature difference over 2 C** (infrared thermometry) vs contralateral foot.
- **Radiographs (weight-bearing): fragmentation, dislocation, rocker-bottom (loss of calcaneal pitch, negative lateral talo-first metatarsal angle)**, **lateral column collapse**.
- **MRI** (stage 0: marrow oedema, microfractures).
- **Charcot vs osteomyelitis (key distinction):**

Feature	Charcot	Osteomyelitis
Location	**Midfoot (joints, periarticular)**, multiple joints	**Forefoot or calcaneus (adjacent to ulcers)**; usually single bone
Ulcer	**Often absent** (or over a bony prominence)	**Ulcer contiguous with the bone; probe to bone positive**
Erythema	**Resolves with elevation**	Persists

Feature	Charcot	Osteomyelitis
MRI	**Subchondral/periarticular oedema; preserved fat; "ghost sign" absent**	**Diffuse marrow replacement (low T1), sinus tract, abscess; "ghost sign" (bone disappears on T1 and reappears on post-contrast)**
Nuclear medicine	**Labelled leucocyte scan: mild uptake** (marrow)	**Intense uptake on labelled leucocyte scan** (combined with marrow scan)
Gold standard		**Bone biopsy**

11. Management

Acute phase (stage 0/I): non-operative is the mainstay

- **Immobilisation and offloading: total contact cast (TCC)** (**changed weekly initially, then every 1-2 weeks**; **non-weight bearing or protected weight bearing**) until **temperature difference falls under 2 C** (stage II/III: **about 3-6 months or longer**).
- **Then: Charcot restraint orthotic walker (CROW)** (**about 6-12 months**), then **custom footwear/AFO** (lifelong).
- **Bisphosphonates/calcitonin: no proven benefit on outcomes** (may reduce markers/temperature; not routine).
- **Monitor the contralateral foot** (**bilateral involvement in about 25-30%**).

Surgery

- **Indications: unstable/unbraceable deformity**, **recurrent/non-healing ulceration over bony prominences despite offloading**, **osteomyelitis**, **progressive deformity**, **pain/instability (ankle Charcot)**.
- **Timing: traditionally in the coalescence/consolidation (stage II-III) phase**; **early surgery in stage I** increasingly accepted for unstable patterns (ankle) with **"superconstruct" fixation**.
- **Procedures:**

  - **Exostectomy** (**plantar bony prominence: medial column (medial cuneiform) prominences do better than lateral column (cuboid)**) for **stable deformities with ulcers**,
  - **Achilles lengthening/gastrocnemius recession** (reduces midfoot load; corrects equinus driving collapse),
  - **Reconstruction (realignment arthrodesis):**
  - **Midfoot: medial column beam (solid bolt through the first metatarsal into the talus), lateral column beams, plantar plating (tension side)**,
  - **Hindfoot/ankle: tibiotalocalcaneal (TTC) arthrodesis with a retrograde IM nail**, **circular (Ilizarov) external fixation** (**infection/osteomyelitis, poor bone, ulcers**),
  - **"Superconstruct" principles (Sammarco): extend fusion beyond the zone of injury to include unaffected joints**, **bony resection to shorten and decompress soft tissues**, **strongest device tolerated by soft tissue**, **devices applied in the position of maximum mechanical function (plantar plating)**,
  - **Two-stage management with infection** (**resection of infected bone, antibiotic spacer, external fixation, then definitive fusion**),
  - **Amputation (BKA)** for **recurrent ulceration/infection, failed reconstruction** (**can give better function than repeated salvage attempts**).
- **Complications: non-union (fibrous union may be stable and acceptable)**, **infection**, **hardware failure**, **recurrence**, **contralateral Charcot (from increased load)**, **amputation (about 5-15%)**, **mortality**.

> Exam pearl: **Neuropathy (10-g monofilament) causes most ulcers; ABI falsely high in calcified vessels: use toe pressures/TBI/TcPO2. Wagner 0-5 (3 = abscess/osteomyelitis, 4 forefoot gangrene); UT grade + stage (stage predicts amputation). Probe-to-bone; MRI best imaging; bone biopsy gold standard.**

Exam pearl: **TCC gold standard offloading (not with infection/ischaemia). Achilles lengthening reduces forefoot ulcer recurrence (transfer heel ulcers). BKA energy cost much lower than AKA. Healing: ABI over 0.45, toe pressure over 30-40, TcPO2 over 30-40, albumin over 3.0, lymphocytes over 1,500.**

Exam pearl: **Charcot: hot swollen neuropathic foot, temp difference over 2 C, erythema resolves with elevation. Eichenholtz I fragmentation, II coalescence, III reconstruction (Shibata 0 = MRI oedema only). Midfoot (Lisfranc) commonest; ankle worst. Treat with TCC until temperature equalises, then CROW. Surgery: unstable/ulcer/infection; superconstruct; medial column beam; TTC nail/Ilizarov.**

Talus and calcaneus

Part A: Talus fractures

1. Anatomy

- **About 60% of the talus is covered by articular cartilage**; **no muscle or tendon attachments**; regions: **head, neck, body (dome), lateral process, posterior process (medial and lateral tubercles; FHL groove between them)**.
- **Blood supply (retrograde to the body, like the scaphoid):**
  - **Artery of the tarsal canal (from the posterior tibial artery): the main supply to the talar body (about 50%)**,
  - **Deltoid branch** (from the artery of the tarsal canal: **supplies the medial body; may be the only remaining supply after neck fractures: preserve the deltoid ligament**),
  - **Artery of the tarsal sinus (from the peroneal and dorsalis pedis arteries)**: lateral body,
  - **Dorsalis pedis branches**: head and neck (superior neck).
  - **Anastomosis of the tarsal canal and sinus arteries beneath the neck (tarsal sling)**.
- **Neck fractures disrupt intraosseous supply; dislocation of the subtalar, ankle and talonavicular joints sequentially destroys the remaining vessels**.

2. Talar neck fractures

- **Mechanism: forced dorsiflexion (axial load): "aviator's astragalus"** (neck impacts the anterior tibia); **high-energy (RTA, falls)**; **about 50% of talus fractures**.
- **Hawkins classification (modified by Canale and Kelly; Vallier subdivision of II):**

Type	Description	AVN risk (approximate)
I	Undisplaced neck fracture	0-15%
II	Displaced + subtalar subluxation/dislocation (IIA subtalar subluxed; IIB subtalar dislocated (Vallier: IIB higher AVN))	20-50%
III	Displaced + subtalar AND ankle (tibiotalar) dislocation (body extruded, usually posteromedially)	50-100% (about 70-90%)
IV	III + talonavicular subluxation/dislocation	Up to about 100%

- **Imaging:** radiographs (**Canale view: 15-degree pronated, 75-degree cephalad-tilted AP of the foot: best view of the talar neck**), **CT (essential: comminution, displacement)**.
- **Management:**
  - **Type I (truly undisplaced, confirmed on CT): non-weight-bearing cast 6-12 weeks** (or fixation to allow early motion).
  - **Displaced: urgent reduction of dislocations (emergency: skin at risk, especially with extruded body)**; **definitive ORIF can be delayed until soft tissues allow** (**timing of fixation does not influence AVN rates (Vallier, Lindvall); the reduction of dislocations is the emergency**).
  - **ORIF via dual anteromedial + anterolateral approaches** (**accurate assessment of reduction: avoid varus malreduction (medial comminution)**), **screws (posterior-to-anterior screws biomechanically stronger) ± mini-fragment plates (lateral/medial bridge plates for comminution)**; **medial malleolar osteotomy** for body access (**preserve the deltoid ligament: deltoid branch**).
- **Complications:**
  - **AVN** (**Hawkins sign: subchondral lucency (radiolucent band) in the talar dome on the AP ankle at 6-8 weeks = revascularisation (bone resorption implies blood supply): good prognostic sign; absence does not guarantee AVN**), **MRI**; **AVN does not always progress to**

collapse; **creeping substitution**; **protected weight bearing debated**.

- **Post-traumatic arthritis (subtalar commonest: about 50-100%)**, **ankle arthritis**,
- **Malunion (varus malunion commonest: medial comminution; leads to hindfoot varus, lateral column overload, reduced subtalar eversion)**,
- **Non-union (about 5-10%)**, **infection, skin necrosis (open fractures: about 20-25% of neck fractures are open)**.
- **Salvage: subtalar fusion, TTC/Blair fusion (tibiotalar fusion with anterior sliding tibial graft into the talar neck/head: for AVN of the body), talectomy (historical)**.

3. Talar body, process and head fractures

- **Talar body fractures** (**intra-articular; ankle and subtalar**): **ORIF often needs malleolar osteotomy**; **AVN and arthritis common**.
- **Lateral process fractures ("snowboarder's fracture")**: **dorsiflexion + inversion/eversion with axial load**; **missed as ankle sprains (pain just anterior and inferior to the tip of the lateral malleolus)**: **CT**; **undisplaced/small: cast**; **large (over 1 cm) displaced: ORIF**; **comminuted: excision**.
- **Posterior process fractures:**
  - **Lateral tubercle (Shepherd fracture)**: **forced plantarflexion**; differentiate from **os trigonum**,
  - **Medial tubercle (Cedell fracture)**: **pronation-dorsiflexion (avulsion of the posterior tibiotalar deltoid)**.
  - **Non-weight-bearing cast; excision of symptomatic non-union fragments.**
- **Talar head fractures**: rare; **compression injuries**; **ORIF or excision of small fragments**.
- **Osteochondral lesions** (see ankle notes).

4. Subtalar and total talar dislocations

- **Subtalar (peritalar) dislocation: simultaneous dislocation of the talocalcaneal and talonavicular joints** with an intact tibiotalar joint.
  - **Medial (about 80%): "acquired clubfoot" appearance (inversion)**; **"basketball foot"**; **irreducible by: talar head buttonholed through the extensor digitorum brevis, extensor retinaculum or talonavicular capsule, peroneal tendons**.
  - **Lateral (about 20%): higher energy, worse prognosis; "acquired flatfoot"**; **irreducible by: tibialis posterior tendon (interposed: commonest block), FHL, FDL**.
  - **Urgent closed reduction (knee flexed to relax the gastrocnemius, accentuate deformity, then reverse)**, **open reduction if irreducible**; **CT after reduction** (associated fractures in about 50%+); **cast about 4-6 weeks**; **subtalar stiffness common**.
- **Total talar extrusion (dislocation)**: **open injury usually**: **reimplant the talus after thorough debridement** (lower complication rates than talectomy); **AVN almost certain** but reimplantation preserves height and function.

Part B: Calcaneus fractures

5. Anatomy and mechanism

- **Commonest tarsal bone fractured (about 60% of tarsal fractures)**; **intra-articular in about 75%**.
- **Articular facets: posterior (largest, main weight-bearing)**, **middle (on the sustentaculum tali)**, **anterior**; **sustentaculum tali (constant fragment: attached to the talus by the interosseous and deltoid ligaments; FHL runs beneath it)**; **anterior process** (calcaneocuboid).
- **Mechanism: axial load (fall from height, RTA)**: the **lateral process of the talus acts as a wedge**, creating a **primary fracture line** (anterolateral to posteromedial: **separating the sustentacular (superomedial) fragment from the tuberosity (superolateral)**) and

secondary fracture lines (**tongue-type vs joint-depression type**: Essex-Lopresti).

- **Associated injuries (about 10% lumbar spine fractures (L1), bilateral calcaneal fractures in about 5-10%, other lower limb fractures)**: **"lover's fracture"/"don Juan" fracture pattern**; **compartment syndrome of the foot (about 10%: calcaneal compartment)**: **claw toe deformity late**.

6. Imaging and measurements

- **Lateral radiograph:**
  - **Böhler angle** (**normal 20-40 degrees**; **reduced/negative = depressed posterior facet: severity indicator, prognostic**),
  - **Critical angle of Gissane** (**normal 95-105 degrees (100-130 in some texts)**; **increased** with posterior facet depression),
  - **Double density sign** (depressed posterior facet fragment).
- **Axial (Harris) view: varus, widening, shortening of the tuberosity**; **AP foot**: CC joint involvement; **Broden views**: posterior facet.
- **CT (essential): Sanders classification (coronal images through the widest part of the posterior facet)**:

Type	Description	Prognosis
I	**Undisplaced (under 2 mm)** posterior facet fracture (any number of lines)	**Non-operative**
II	**Two-part** fracture of the posterior facet (**IIA lateral, IIB central, IIC medial** fracture line)	**Best results with ORIF**
III	**Three-part** fracture with a **depressed central fragment** (IIIAB, IIIAC, IIIBC)	Intermediate
IV	**Four or more parts: highly comminuted**	**Poor: primary subtalar arthrodesis considered**

- **Essex-Lopresti classification (lateral radiograph): joint-depression type** (posterior facet separate from the tuberosity) vs **tongue-type** (**secondary fracture line exits posteriorly through the tuberosity: the**

posterior facet is attached to the tongue fragment: **posterior skin at risk: urgent reduction/percutaneous fixation (Essex-Lopresti manoeuvre)**).

7. Management

- **Soft tissues: the key determinant of timing**: **wait for the "wrinkle sign" (return of skin wrinkles: about 7-14 days)**; **fracture blisters (blood-filled = deeper injury: avoid incisions through them)**; elevation, foot pump.
- **Emergencies: open fractures** (often medial wound), **compartment syndrome**, **tongue-type fractures with posterior skin tenting** (**reduce within 24 hours**), **fracture-dislocations**.
- **Non-operative: Sanders I**, **extra-articular fractures (minimal displacement)**, **elderly, smokers, diabetes, PVD, poor compliance, severe soft-tissue injury**, non-ambulators: **early ROM, non-weight bearing 6-12 weeks**.
- **Operative (displaced intra-articular, Sanders II-III; tuberosity displacement):**
  - **Extensile lateral L-shaped approach** (**full-thickness flap; lateral calcaneal artery supplies the corner: wound complications/dehiscence in about 10-25%**; **sural nerve**): **ORIF with lateral plate**: **historical standard**,
  - **Sinus tarsi (minimally invasive) approach** (**fewer wound complications (about 2-5%)**, similar reduction for Sanders II-III) **(now preferred)**,
  - **Percutaneous reduction and screw fixation** (tongue-type, Sanders II),
  - **Primary subtalar arthrodesis (with ORIF of the body shape)** for **Sanders IV**.
- **Goals: restore the posterior facet congruity, Böhler angle (height), calcaneal width, length, hindfoot alignment (correct varus)**.
- **Evidence:**

- **Buckley (Canadian RCT 2002): no overall difference between operative and non-operative treatment**; **surgery benefited: women, younger patients, non-workers' compensation, lighter work, Böhler angle 0-14 degrees, Sanders II**; **non-operative patients 5.5x more likely to need late subtalar fusion**.
- **UK HeFT trial (Griffin 2014): no benefit of ORIF (extensile lateral) at 2 years; more complications in the surgical group**.
- **Smokers, diabetics, workers' compensation: worse outcomes and higher wound complications (smoking: about 3x wound complications)**.

8. Complications

- **Wound dehiscence/infection** (**extensile lateral approach; smokers, diabetes, open fractures, delayed surgery under 7 days with swelling**), **sural nerve injury**.
- **Subtalar arthritis (commonest late problem)**: **subtalar arthrodesis (in situ)**; with **loss of height (horizontal talus, anterior ankle impingement, loss of dorsiflexion)**: **subtalar bone-block distraction arthrodesis (restores talar declination)**.
- **Lateral wall blowout**: **calcaneofibular (subfibular) impingement**, **peroneal tendon impingement/dislocation (about 30%)** and **stenosis**: **lateral wall exostectomy, peroneal tendon decompression**.
- **Malunion: widening (heel widening: shoe problems), shortening, varus hindfoot**: **calcaneal osteotomy + subtalar fusion** (Stephens-Sanders classification of malunion: **I lateral wall exostosis, II + subtalar arthritis, III + varus malalignment**).
- **Compartment syndrome** (missed: **claw toes, intrinsic contracture**), **heel pad pain (fat pad damage)**, **CRPS**, **tibial nerve entrapment (tarsal tunnel)**.

9. Extra-articular calcaneal fractures

- **Anterior process fractures** (**avulsion of the bifurcate ligament with inversion/plantarflexion: mimics sprain**; **small: boot/cast; large (over**

25% CC joint) displaced: ORIF; symptomatic non-union: excision).

- **Calcaneal tuberosity avulsion fractures** (**Achilles avulsion; diabetics, osteoporosis: "beak fracture"**): **posterior skin at risk: emergency reduction and fixation**.
- **Sustentaculum tali fractures** (isolated; **FHL entrapment**): ORIF if displaced.
- **Body fractures not involving the subtalar joint**: usually non-operative unless significant deformity.
- **Calcaneal stress fractures** (**military recruits, runners: sclerotic line parallel to the posterior border**): rest.
- **Paediatric calcaneus fractures**: **more extra-articular, remodel**: mostly non-operative.

Part C: Midfoot injuries (brief link)

- **Lisfranc (tarsometatarsal) injury: Lisfranc ligament: from the medial cuneiform to the base of the second metatarsal (plantar band strongest)**; **plantar ecchymosis sign**, **"fleck sign" (avulsion at the base of the second metatarsal)**; **weight-bearing radiographs: diastasis between the first and second metatarsal bases (over 2 mm)**, **medial borders of the second metatarsal and middle cuneiform should align (AP)**, **fourth metatarsal and cuboid (oblique)**; **Myerson classification (A total incongruity, B partial (B1 medial, B2 lateral), C divergent)**; **anatomic reduction essential**: **ORIF (bridge plating/screws) vs primary arthrodesis (medial 2-3 TMT joints) for purely ligamentous injuries (Ly and Coetzee: primary fusion gave better outcomes and fewer reoperations)**; **lateral column (4th/5th TMT) never fused (mobility essential): K-wire temporarily**.
- **Navicular fractures** (**body: Sangeorzan types; stress fractures: central third**), **cuboid "nutcracker" fractures** (forced abduction: **restore lateral column length**).
- **Fifth metatarsal base fractures: zone 1 (tuberosity avulsion: peroneus brevis/lateral band of plantar fascia: symptomatic treatment)**, **zone 2 (Jones fracture: metaphyseal-diaphyseal junction**

(4th-5th intermetatarsal joint): watershed: non-weight-bearing cast 6-8 weeks or intramedullary screw in athletes), zone 3 (proximal diaphyseal stress fracture: screw fixation often).

Exam pearl: **Talus: artery of the tarsal canal (posterior tibial) = main body supply; deltoid branch: preserve the deltoid. Hawkins I-IV AVN rising to near 100%; Hawkins sign (subchondral lucency at 6-8 weeks) = revascularisation. Canale view for the neck. Reduce dislocations urgently; ORIF timing doesn't change AVN. Varus malunion from medial comminution. Subtalar arthritis commonest.**

Exam pearl: **Lateral process = snowboarder's fracture (mimics sprain). Medial subtalar dislocation blocked by EDB/talar head buttonhole; lateral blocked by tibialis posterior.**

Exam pearl: **Calcaneus: Böhler 20-40 (reduced), Gissane 95-105 (increased). Sanders CT (coronal, posterior facet): I non-op, II best ORIF results, IV consider primary subtalar fusion. Tongue-type with skin tenting = urgent. Wait for wrinkle sign; sinus tarsi approach fewer wound problems. Buckley: no overall difference (women, young, non-compensation benefit); HeFT: no benefit. Late: subtalar arthritis, lateral wall impingement/peroneals; loss of height = bone-block distraction fusion. Check L-spine (10%).**

Exam pearl: **Lisfranc: plantar ecchymosis, fleck sign, 1-2 MT base diastasis; ligamentous = primary medial column fusion; never fuse 4th/5th TMT. Fifth MT: zone 1 avulsion (symptomatic), zone 2 Jones (watershed: cast NWB or IM screw).**